Decision and Risk Analysis

Lecture 2: **Bayes' Rule**

DAEN 427/ISEN 427/ISEN 627

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Fake News Detection on Social Media: A Data Mining Perspective

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ABSTRACT

Social media for news consumption is a double-edged sword. On the one hand, its low cost, easy access, and rapid dissemination of information lead people to seek out and consume news from social media. On the other hand, it enables the wide spread of "fake news", i.e., low quality news with intentionally false information. The extensive spread of fake news has the potential for extremely negative impacts on individuals and society. Therefore, fake news detection on social media has recently become an emerging research that is attracting tremendous attention. Fake news detection on social media presents unique characteristics and challenges that make existing detection algorithms from traditional news media ineffective or not applicable. First, fake news is intentionally written to mislead readers to believe false information, which makes it difficult and nontrivial to detect based on news content; therefore, we need to include auxiliary information, such as user social engagements on social media, to help make a determination. Second, exploiting this auxiliary information is challenging in and of itself as users' social engagements with fake news produce data that is big, incomplete, unstructured, and noisy. Because the issue of fake news detection on social media is both challenging and relevant, we conducted this survey to further facilitate research on the problem. In this survey, we present a comprehensive review of detecting fake news on social media, including fake news characterizations on psychology and social theories, existing algorithms from a data mining perspective, evaluation metrics and representative datasets. We also discuss related research areas, open problems, and future research directions for fake news detection on social media.

1. INTRODUCTION

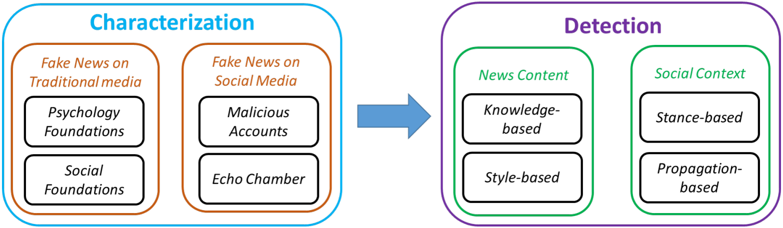
As an increasing amount of our lives is spent interacting online through social media platforms, more and more people tend to seek out and consume news from social media rather than traditional news organizations. The reasons for this change in consumption behaviors are inherent in the nature of these social media platforms: (i) it is often more timely and less expensive to consume news on social media compared with traditional news media, such as newspapers or television; and (ii) it is easier to further share, comment on, and discuss the news with friends or other readers on social media. For example, 62 percent of U.S. adults get news on social media in 2016, while in 2012, only 49 percent reported seeing news on social media[1]. It was also found that social media now outperforms television as the major news source[2]. Despite the advantages provided by social media, the quality of news on social media is lower than traditional news organizations. However, because it is cheap to provide news online and much faster and easier to disseminate through social media, large volumes of fake news, i.e., those news articles with intentionally false information, are produced online for a variety of purposes, such as financial and political gain. It was estimated that over 1 million tweets are related to fake news "Pizzagate"[3] by the end of the presidential election. Given the prevalence of this new phenomenon, "Fake news" was even named the word of the year by the Macquarie dictionary in 2016.

The extensive spread of fake news can have a serious negative impact on individuals and society. First, fake news can break the authenticity balance of the news ecosystem. For example, it is evident that the most popular fake news was even more widely spread on Facebook than the most popular authentic mainstream news during the U.S. 2016 president election[4]. Second, fake news intentionally persuades consumers to accept biased or false beliefs. Fake news is usually manipulated by propagandists to convey political messages or influence. For example, some report shows that Russia has created fake accounts and social bots to spread false stories[5]. Third, fake news changes the way people interpret and respond to real news. For example, some fake news was just created to trigger people's distrust and make them confused, impeding their abilities to differentiate what is true from what is not[6]. To help mitigate the negative effects caused by fake news–both to benefit the public and the news ecosystem–It's critical that we develop methods to automatically detect fake news on social media.

[1]http://www.journalism.org/2016/05/26/news-use-across-social-media-platforms-2016/
[2]http://www.bbc.com/news/uk-36528256
[3]https://en.wikipedia.org/wiki/Pizzagate_conspiracy_theory
[4]https://www.buzzfeed.com/craigsilverman/viral-fake-election-news-outperformed-real-news-on-facebook?utm_term=.nrg0WA1VP0#.gjJyKapW5y
[5]http://time.com/4783932/inside-russia-social-media-war-america/
[6]https://www.nytimes.com/2016/11/28/opinion/fake-news-and-the-internet-shell-game.html?_r=0

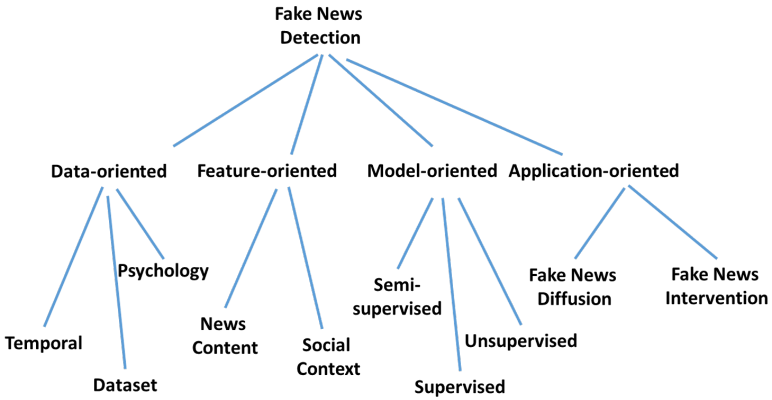
https://arxiv.org/abs/1708.01967 for details.

Framework



https://arxiv.org/abs/1708.01967 for details.

Future Directions



https://arxiv.org/abs/1708.01967 for details.

We're Using R Instead of Python

Python

- "Can do everything" ... installs n packages, breaks after update
- Stack Overflow becomes your TA

R $\rightarrow$ `https://cran.r-project.org`

- Literally for statistics; `lm(y ~ x)` and move on with your life
- Excellent, open-source, keeps us focused on data analysis,
  ... and I enjoy sleeping at night!

What I Need From You: learn concepts, analyze data, pass the class!
What I Do Not Need: a 2-hour debate about virtual environments.

Building a Bayesian Model for Events

Prior Probability Model

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Dataset

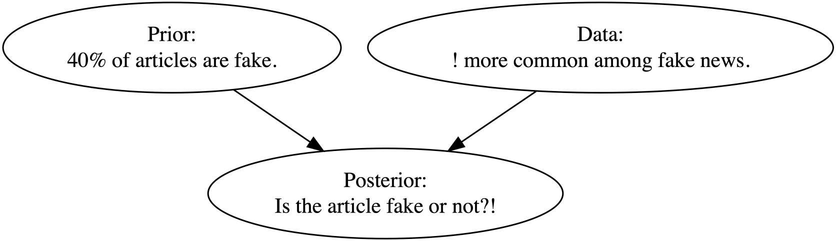
```r
# Load packages
library(bayesrules)
library(tidyverse)
library(janitor)

# Import article data
data(fake_news)

fake_news %>%
  tabyl(type) %>%
  adorn_totals("row")
#  type   n percent
#  fake  60     0.4
#  real  90     0.6
# Total 150     1.0
```

- 150 news articles: 60 fake, 90 real
- $P(\text{Fake}) = \frac{60}{150} = 0.40$. Before looking at the headline, the prior probability that an article is fake is 40%.

Bayesian reasoning to estimate whether a news article is fake:

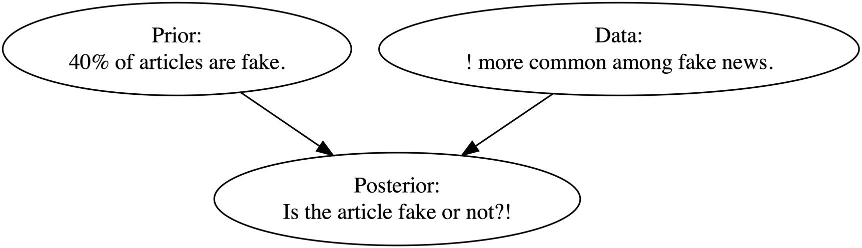


```r
# Transposed version of head()
# ... makes possible to see all columns in a data frame
glimpse(fake_news)
```

Observed headline pattern:

```r
# Tabulate exclamation usage and article type
fake_news %>%
  tabyl(title_has_excl, type) %>%
  adorn_totals("row")
# title_has_excl fake real
#          FALSE   44   88
#           TRUE   16    2
#          Total   60   90
```

Exclamation Points as Evidence



Title Has "!"	Fake	Real	Total
Yes	16	2	18
No	44	88	132
Total	60	90	150

Exclamation Points as Evidence

Title Has "!"	Fake	Real	Total
Yes	16	2	18
No	44	88	132
Total	60	90	150

$$P(! \mid \text{Fake}) = \frac{16}{60} \approx 0.267,\ P(! \mid \text{Real}) = \frac{2}{90} \approx 0.022$$

An exclamation point increases the probability that an article is fake.

$$P(\text{Fake} \mid !) = \frac{P(\text{Fake})\, P(! \mid \text{Fake})}{P(!)}$$

Conditional Probability & Likelihood

Conditional vs Unconditional Probability

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Independent Events

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Probability vs Likelihood

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Normalizing Constants

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Calculating Joint and Conditional Probabilities

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Posterior Probability Model via Bayes' Rule

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Posterior Simulation

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Building a Bayesian Model for Random Variables

Prior Probability Model

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Discrete Probability Model

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The Binomial Data Model

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Conditional Probability Model of Data Y

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The Binomial Model

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The Binomial Likelihood Function

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Probability Mass Functions vs Likelihood Functions

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Normalizing Constant

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Posterior Probability Model

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Bayes' Rule for Variables

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Posterior Shortcut

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Proportionality

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Posterior Simulation

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Quiz 1

A medical test correctly detects a disease 95% of the time when the disease is present. This value represents:

a. Prior probability
b. Posterior probability
c. False positive rate
d. Likelihood

Answer: d. Likelihood

The statement describes: $P(\text{Test positive} \mid \text{Disease present}) = 0.95$

This is the probability of observing the test result given that the disease is present, which represents the likelihood (also called **sensitivity** or **true positive rate**).

- **Prior**: Probability of disease before observing the test result.
- **Posterior**: Probability of disease after observing the test result.
- **False positive rate**: Probability that the test is positive when the disease is absent.

Quiz 2

A university uses an AI system to detect plagiarism.

- 5% of assignments are actually plagiarized.
- The AI flags plagiarized assignments 92% of the time.
- The AI incorrectly flags honest assignments 8% of the time.

If an assignment is flagged, what is the probability it was actually plagiarized? Briefly discuss whether the AI system seems reliable.

Answer:

$$P(P \mid F) = \frac{0.92 \cdot 0.05}{0.92 \cdot 0.05 + 0.08 \cdot 0.95} \approx 0.377$$

Approximately 37.7% chance that it is actually plagiarism.
The system catches most plagiarism cases, but because plagiarism is rare, many flagged assignments are false positives.

Quiz 3

Let's suppose that you are an air traveler along with 99 other passengers from your flight, each of whom, like you, checked one item of luggage.

A recording piped through the speakers reminds you that: *"Many bags look alike. Please check your bag carefully before exiting the terminal."*

Indeed, your bag is one of the most popular models on the market, a medium-sized black rectangular case **used by 5% of all travelers**.

Given your distance from the luggage chute, you can discern only the shape, size, and color of the bags that emerge, not any detailed markings on the bags.

Quiz 3, cont.

a. Let's suppose that the **first bag** from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain (i.e., calculate the posterior probability that the 1st bag is yours).

b. Now suppose that we continue to wait for our bag to appear, failing to see it among the first 98 bags to enter the carousel (i.e., only 2 bags are left to come off the airplane). Calculate the posterior probability that the **99th** bag, which also matches ours in shape, size, and color, is our own.

Answer: a. Let Y = first bag is yours, and M = first bag matches.

$$P(Y) = \frac{1}{100}$$
$$\mathcal{L}(Y^c \mid M) = P(M \mid Y^c) = 0.05$$
$$\mathcal{L}(Y \mid M) = P(M \mid Y) = 1$$

$$P(Y \mid M) = \frac{P(M \mid Y)P(Y)}{P(M)} = \frac{P(M \mid Y)P(Y)}{P(M \mid Y)P(Y) + P(M \mid Y^c)P(Y^c)} =$$
$$= \frac{1 \cdot 0.01}{1 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.01}{0.0595} \approx 0.168$$

So even though the first bag matches, it is probably not yours.

Answer: b.

$$P(Y \mid M) = \frac{1 \cdot \frac{1}{\text{total of bags remaining}}}{1 \cdot \frac{1}{\text{total of bags remaining}} + 0.05 \cdot \frac{\text{total of bags remaining}-1}{\text{total of bags remaining}}}$$

(if 2 bags remain)

$$= \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + 0.05 \cdot \frac{1}{2}} = \frac{1}{1 + 0.05} \approx 0.9524$$

There is a 95% chance that the 99th bag is yours, given that it looks like yours.

Or, it is **19 times more likely** to be yours than not yours $^{(0.95/0.05=19)}$.

Homework

- Submit your answers before the next class.
- You may discuss with peers but submissions must be individual.
- Responses may be shared anonymously during class discussion!

Question 1

Define the following events for a resident of a fictional town as:

- $A =$ drives 10 miles per hour above the speed limit,
- $B =$ gets a speeding ticket,
- $C =$ took statistics at the local college,
- $D =$ has used R,
- $E =$ likes the music of Prince, and
- $F =$ is a Texan.

Several facts about these events are listed next.

Question 1, cont.

Specify each of these facts using probability notation, paying special attention to whether it's a marginal, conditional, or joint probability.

- 73% of people that drive 10 miles per hour above the speed limit get a speeding ticket.
- 20% of residents drive 10 miles per hour above the speed limit.
- 15% of residents have used R.
- 91% of statistics students at the local college have used R.
- 38% of residents are Texans that like the music of Prince.
- 95% of the Texans like the music of Prince.

Question 2

Edward is trying to prove to Bella that vampires exist. Bella thinks there is a 0.05 probability that vampires exist. She also believes that the probability that someone can sparkle like a diamond if vampires exist is 0.7, and the probability that someone can sparkle like a diamond if vampires don't exist is 0.03. Edward then goes into a meadow and shows Bella that he can sparkle like a diamond. Given that Edward sparkled like a diamond, what is the probability that vampires exist?

Question 3

The probability that Sandra will like a restaurant is 0.7. Among the restaurants that she likes, 20% have five stars on Yelp, 50% have four stars, and 30% have fewer than four stars. What other information do we need if we want to find the posterior probability that Sandra likes a restaurant given that it has fewer than four stars on Yelp?

Question 4

Robin applies for six equally competitive data science internships. He has the following prior model for his chances of getting into any given internship, π:

π	0.3	0.4	0.5	Total
$f(\pi)$	0.25	0.60	0.15	1

(a) Let Y be the number of internship offers that Robin gets. Specify the model for the dependence of Y on π and the corresponding pmf, $f(y \mid \pi)$.

(b) Robin got some pretty amazing news. He was offered four of the six internships! How likely would this be if $\pi = 0.3$?

(c) Construct the posterior model of π in light of Robin's internship news.

Question 5

Whether you like it or not, cats have taken over the internet. Joining the craze, Michael has written an algorithm to detect cat images. It correctly identifies 80% of cat images as cats, but falsely identifies 50% of non-cat images as cats. Michael tests her algorithm with a new set of images, 8% of which are cats. What's the probability that an image is actually a cat if the algorithm identifies it as a cat? Answer this question by simulating data for 10,000 images.